Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Theory

Data and methods

Hypothesis 1a
Hypothesis 1b
Hypothesis 2
Hypothesis 3

Some studies

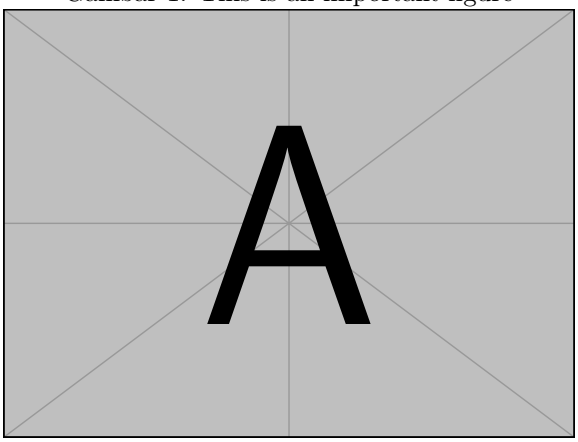
The most cited papers in AMJ according to their website are [32]doi:10.5465/amj.2007.24160888, doi:10.5465/amj.2007.28166119, doi:10.5465/amj.2013.4001, and doi:10.5465/amj.2016.4007.[1]

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Figure 1 is added directly via the `figure` and `minipage` environments.

Gambar 1: This is an important figure



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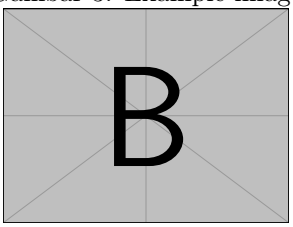
Gambar 2: *

Notes: B is a letter in the alphabet. It comes right after the letter A, and is preceded by the letter C.

[1]This is an example of a footnote. See `https://journals.aom.org/journal/amj`.

I also wrapped these in a new environment, `add[l]figure` (`l` for landscape) , which takes four arguments: (1) width, (2) image file name, (3) caption (with optional label), and (4) figure note.

Gambar 3: Example image



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Gambar 4: *

Notes: This is an example image.

Tabel 1: This is a description of some data

A	B	C
1	2	3
x	y	z

Let's see if it works. Further calculations are presented in Table A1 given in Appendix A. Similarly, supplemental analyses, such as Table B1, are given in Appendix B. The `endfloat` package is disabled for appendices, that is, tables and figures appear within text.

[b]

Tabel 2: Treatment Effects With New Strategy

	Avg Test Score			Avg Passing Rate			Avg Participation Rate		
2-4 5-7 8-10 VARIABLES	Male	Female	All	Male	Female	All	Male	Female	All
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treatment	-26.21	-22.18	34.08*	35.83*	-59.34*	17.13	-26.21	9.84	-8.46
	[-2.02]	[-0.50]	[2.42]	[2.23]	[-2.46]	[1.22]	[-2.02]	[1.24]	[-0.78]
	(13.00)	(44.77)	(14.06)	(16.04)	(24.12)	(14.03)	(13.00)	(7.93)	(10.82)
Post	-26.50**	-25.47	-11.53	40.49***	52.71***	52.68***	-26.50**	2.28	-16.49*
	[-3.52]	[-1.35]	[-1.34]	[4.36]	[5.18]	[6.15]	[-3.52]	[0.68]	[-2.49]
	(7.52)	(18.90)	(8.59)	(9.28)	(10.18)	(8.57)	(7.52)	(3.35)	(6.61)
DiD	29.65*	31.22	5.33	9.17	141.80**	11.56	29.65*	-15.08	1.07
	[2.13]	[0.39]	[0.32]	[0.53]	[3.25]	[0.70]	[2.13]	[-1.05]	[0.08]
	(13.89)	(80.90)	(16.45)	(17.14)	(43.58)	(16.42)	(13.89)	(14.34)	(12.67)
Constant	42.02	271.27*	222.43***	65.29*	-23.66	-38.16	42.02	23.55	-46.42
	[1.87]	[2.58]	[6.20]	[2.36]	[-0.42]	[-1.07]	[1.87]	[1.26]	[-1.68]
	(22.46)	(105.12)	(35.85)	(27.71)	(56.63)	(35.78)	(22.46)	(18.63)	(27.59)
Obs	39	38	47	39	38.00	47	39	38	47

[para,flushleft]

Notes: [para,flushleft]

Source: `https://tex.stackexchange.com/questions/513415/making-table-notes-aligned-with-the-table-width`

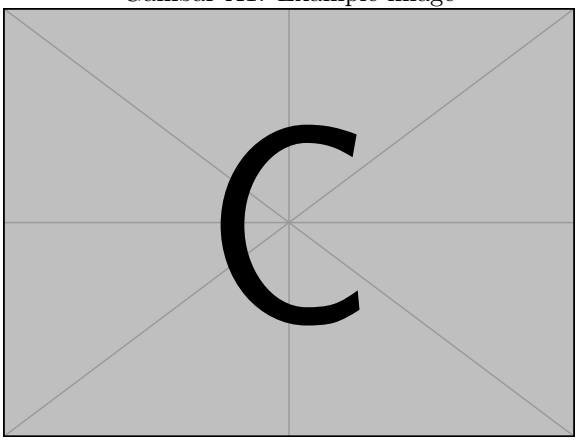
† $p < 0.10$
* $p < 0.05$
** $p < 0.01$
*** $p < 0.001$

References

APPENDIX :0.5em

A Further Calculations

Gambar A1: Example image



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Gambar A2: *

Notes: This is an example image.

Tabel A1: This is a description of some other data

A	B	C
1	2	3
x	y	z

B Supplemental Analyses

Tabel B1: This is a description of some other data

A	B	C
1	2	3
x	y	z

[b]

Tabel B2: This is a description of some other data

Uppercase letter:	A	B	C
Number:	1	2	3
Lowercase letter:	x	y	z

[para,flushleft]

Notes: There are many interesting things to say about this table. [para,flushleft]

Source: Bureau of Labor Statistics, April 2022.